```
      points.insert(node.point)
    if compare(target, node) < 0
then                                        #6
      closeBranch ← node.left
      farBranch ← node.right
    else

    #7
      closeBranch ← node.right
      farBranch ← node.left
    points.insertAll(pointsInSphere(closeBranch, center,
radius))   #8
   if splitDistance(target, node) < radius
then                               #9
     points.insertAll(pointsInSphere(farBranch, center,
radius))   #10
   return points

#11
```

#1 Find all the points in the container intersecting a given hyper-sphere. We pass the hyper-sphere as its center and radius.

#2 If node is `null`, we are traversing an empty tree, so there is no point to be added.

#3 Otherwise, we have 3 tasks: check if current node is inside the hyper-sphere, traverse the branch on the same side of the split with respect to the center of the sphere, and check if we can prune the other branch (or traverse it as well). We start by initializing the list of points found within this subtree.